

Multiple Choice (10 marks)

1. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan
2. The sub key length at each round of DES is_____
 - (a) 32
 - (b) 56
 - (c) 48
 - (d) 64
3. The four Primary Security Principles related to messages are
 - (a) Confidentiality, Integrity, Non repudiation and Authentication
 - (b) Confidentiality, Access Control, Integrity, Non repudiation
 - (c) Authentication, Authorization, Availability, Integrity
 - (d) Availability, Authorization, Confidentiality, Integrity
4. Which Nmap scan is does not completely open a TCP connection?
 - (a) SYN stealth scan
 - (b) TCP connect
 - (c) XMAS tree scan
 - (d) ACK scan
5. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
 - (a) Joined
 - (b) Authenticated
 - (c) Submitted
 - (d) Shared

True / False (10 marks)

Answer True or False

6. True or False: In message confidentiality, the transmitted message must be understandable to both the sender and the translator.
7. True or False: The Deep Web comprises content that is not indexed by standard search engines, including databases and private networks.
8. True or False: The digest created by a hash function is typically known as a Message Authentication Control (MAC).
9. True or False: A ping sweep is employed to identify live systems within a network by sending ICMP echo requests.
10. True or False: The IP Network Browser is an effective tool for performing SNMP enumeration in network security assessments.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of the Tor browser in enabling anonymous internet users to access the Tor network. Analyze its significance in the context of cybersecurity and privacy protection.
12. Discuss the role of AddressSanitizer (ASAN) in identifying memory errors during fuzz testing of programs, and analyze how it enhances the debugging process in cybersecurity.

End of Paper